

## SULLIVAN COUNTY AP, NY, USA

WMO: 725145

Lat: 41.701N

Lon: 74.795W

Elev: 428

StdP: 96.29

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 54746

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -17.8      | -15.2 | -23.8                         | 0.5 | -16.8 | -21.3 | 0.6 | -13.4 | 11.9                  | -8.3 | 10.9 | -7.9 | 3.9                   | 290  | 0.529 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.3                   | 30.2            | 21.7 | 28.6 | 20.7 | 27.3 | 19.9 | 22.9                | 27.8 | 21.9 | 26.6 | 21.1 | 25.5 | 3.7                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.3                            | 16.8 | 25.3 | 20.9 | 16.3 | 24.8 | 19.3 | 14.8 | 23.5 | 69.8          | 27.6 | 66.3 | 27.1 | 63.4 | 25.1 | 27.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 10.0              | 8.7  | 7.7 | -22.1                      | 32.6 | 3.2                | 1.4 | -24.4                                              | 33.6 | -26.3      | 34.4 | -28.1      | 35.1 | -30.4      | 36.1 |
|                   |      |     | -21.7                      | 24.2 | 3.5                | 1.0 | -24.2                                              | 24.9 | -26.3      | 25.5 | -28.2      | 26.1 | -30.8      | 26.8 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.3         | -5.1  | -3.7  | 1.0   | 7.5   | 13.4  | 18.0  | 20.6  | 19.8  | 15.7  | 9.6   | 3.2   | -1.8  |
|                                                                        | DBStd       | 10.07       | 6.04  | 5.58  | 5.48  | 4.95  | 4.33  | 3.56  | 2.89  | 2.87  | 3.88  | 4.54  | 4.83  | 4.91  |
|                                                                        | HDD10.0     | 1895        | 468   | 384   | 283   | 105   | 14    | 0     | 0     | 0     | 2     | 64    | 209   | 366   |
|                                                                        | HDD18.3     | 3894        | 726   | 617   | 537   | 327   | 164   | 48    | 11    | 18    | 97    | 271   | 454   | 623   |
|                                                                        | CDD10.0     | 1261        | 1     | 1     | 5     | 29    | 119   | 241   | 329   | 304   | 174   | 52    | 6     | 1     |
|                                                                        | CDD18.3     | 218         | 0     | 0     | 0     | 2     | 11    | 39    | 82    | 64    | 19    | 2     | 0     | 0     |
|                                                                        | CDH23.3     | 1767        | 0     | 0     | 3     | 30    | 144   | 327   | 658   | 460   | 134   | 12    | 0     | 0     |
|                                                                        | CDH26.7     | 417         | 0     | 0     | 1     | 7     | 32    | 75    | 176   | 101   | 26    | 1     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 3.1         | 3.6   | 3.7   | 3.9   | 3.7   | 3.1   | 2.7   | 2.5   | 2.3   | 2.5   | 3.0   | 3.3   | 3.5   |
| (15) Precipitation                                                     | PrecAvg     | 1246        | 93    | 70    | 100   | 97    | 96    | 120   | 112   | 117   | 119   | 118   | 95    | 108   |
|                                                                        | PrecMax     | 1588        | 181   | 197   | 174   | 175   | 171   | 303   | 205   | 223   | 263   | 346   | 139   | 194   |
|                                                                        | PrecMin     | 861         | 34    | 39    | 23    | 31    | 27    | 39    | 32    | 40    | 43    | 21    | 26    | 46    |
|                                                                        | PrecStd     | 183         | 41    | 34    | 35    | 39    | 42    | 65    | 51    | 50    | 64    | 72    | 31    | 41    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 13.4  | 13.8  | 21.0  | 27.4  | 29.8  | 31.3  | 32.4  | 32.1  | 29.1  | 25.1  | 18.8  | 14.0  |
|                                                                        |             | MCWB        | 11.7  | 8.8   | 13.9  | 16.1  | 20.9  | 21.9  | 22.7  | 22.2  | 21.6  | 17.3  | 13.7  | 11.4  |
|                                                                        | 2%          | DB          | 9.1   | 9.8   | 16.2  | 22.7  | 27.3  | 28.7  | 30.8  | 29.0  | 27.2  | 22.2  | 16.0  | 11.0  |
|                                                                        |             | MCWB        | 7.7   | 5.7   | 9.8   | 13.5  | 18.6  | 20.7  | 22.0  | 21.3  | 20.0  | 16.0  | 12.3  | 9.2   |
|                                                                        | 5%          | DB          | 6.0   | 7.0   | 12.6  | 18.9  | 24.1  | 27.2  | 28.6  | 27.5  | 24.7  | 19.4  | 13.0  | 7.8   |
|                                                                        |             | MCWB        | 3.9   | 3.5   | 7.9   | 11.8  | 16.5  | 20.0  | 20.9  | 20.3  | 19.0  | 15.4  | 9.8   | 6.1   |
|                                                                        | 10%         | DB          | 2.8   | 3.8   | 9.0   | 16.3  | 22.1  | 25.1  | 27.2  | 26.2  | 22.6  | 17.5  | 11.2  | 5.8   |
|                                                                        |             | MCWB        | 1.3   | 1.3   | 5.6   | 9.9   | 15.1  | 18.6  | 20.1  | 19.5  | 17.7  | 13.5  | 8.5   | 3.6   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 12.4  | 11.4  | 14.2  | 17.6  | 21.8  | 23.3  | 24.2  | 24.0  | 23.0  | 19.7  | 15.4  | 12.4  |
|                                                                        |             | MCDB        | 13.3  | 12.7  | 19.5  | 24.8  | 27.5  | 28.7  | 30.0  | 28.8  | 27.4  | 21.8  | 17.4  | 13.7  |
|                                                                        | 2%          | WB          | 8.0   | 6.6   | 11.3  | 15.1  | 19.8  | 21.9  | 22.9  | 22.6  | 21.3  | 17.8  | 13.1  | 9.1   |
|                                                                        |             | MCDB        | 8.7   | 9.3   | 14.9  | 20.4  | 25.3  | 26.7  | 28.2  | 27.0  | 24.8  | 20.3  | 15.3  | 10.2  |
|                                                                        | 5%          | WB          | 4.0   | 3.7   | 8.5   | 13.1  | 18.0  | 21.0  | 21.9  | 21.5  | 20.0  | 15.9  | 10.9  | 6.6   |
|                                                                        |             | MCDB        | 5.6   | 5.9   | 12.0  | 18.2  | 22.4  | 25.4  | 26.8  | 25.9  | 23.3  | 18.9  | 12.9  | 7.8   |
|                                                                        | 10%         | WB          | 1.7   | 1.7   | 5.9   | 10.9  | 16.4  | 19.8  | 21.1  | 20.6  | 18.9  | 13.9  | 8.2   | 3.5   |
|                                                                        |             | MCDB        | 2.8   | 4.0   | 9.3   | 15.4  | 20.2  | 23.9  | 25.5  | 24.5  | 21.7  | 17.1  | 10.3  | 5.2   |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.2   | 8.9   | 9.6   | 11.2  | 11.5  | 11.1  | 11.3  | 11.0  | 10.8  | 10.2  | 8.7   | 7.2   |
|                                                                        |             | MCDBR       | 10.8  | 12.2  | 14.7  | 16.3  | 15.4  | 13.5  | 12.9  | 12.3  | 13.0  | 13.6  | 12.1  | 10.0  |
|                                                                        | 5% WB       | MCWBR       | 9.8   | 9.3   | 9.6   | 8.9   | 8.2   | 7.1   | 5.9   | 6.0   | 7.1   | 8.4   | 8.9   | 8.9   |
|                                                                        |             | MCWBR       | 10.3  | 11.1  | 13.6  | 14.4  | 13.7  | 12.0  | 11.2  | 10.7  | 10.7  | 11.4  | 10.5  | 9.2   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.287 | 0.310 | 0.332 | 0.346 | 0.384 | 0.438 | 0.445 | 0.429 | 0.362 | 0.343 | 0.305 | 0.288 |
|                                                                        |             | taud        | 2.409 | 2.350 | 2.395 | 2.419 | 2.338 | 2.210 | 2.202 | 2.239 | 2.405 | 2.470 | 2.531 | 2.497 |
|                                                                        | Ebn at Noon |             | 867   | 896   | 916   | 924   | 892   | 840   | 829   | 832   | 872   | 849   | 844   | 840   |
|                                                                        |             | Edh at Noon | 79    | 98    | 106   | 111   | 124   | 141   | 141   | 132   | 104   | 87    | 70    | 67    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.67  | 2.55  | 3.57  | 4.52  | 5.12  | 5.51  | 5.74  | 5.02  | 4.15  | 2.74  | 1.86  | 1.35  |
|                                                                        | RadStd      |             | 0.12  | 0.28  | 0.30  | 0.49  | 0.52  | 0.52  | 0.35  | 0.31  | 0.47  | 0.31  | 0.20  | 0.15  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (61 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +108    | +58     |

Nomenclature: See separate page